23

Lecture 23

The 5G User Plane

صفحهٔ کاربر نسل پنجم

2G — 3G — 4G — **5G** — 6G

COURSE PART
Part V – Planes and outlook (Lectures 23–26)

DIFFICULTY
Advanced

ESTIMATED STUDY TIME
6–8 hours (3 h reading, 3–5 h exercises)

PREREQUISITES
Lectures 6, 13, 16, 19 and 22

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Original lecture notes for the course *Advanced Communication Networks*. The lecture sequence follows the course syllabus (26 lectures, 2G to 6G). Equations use identical notation in the English and Persian editions. Companion edition: Persian (`lecture-23-fa.pdf`).

1 Learning objectives

After studying this lecture you should be able to:

- **LO1** **Draw** the 5G user-plane protocol stack from UE to data network, including the gNB's CU/DU split.
- **LO2** **Explain** the roles of SDAP, PDCP, RLC and MAC, and of GTP-U on F1-U, N3 and N9.
- **LO3** **Compare** RAN functional splits and **estimate** their fronthaul requirements.
- **LO4** **Describe** UPF packet processing with PFCP rules and its implementation options.
- **LO5** **Compute** header overhead, latency budgets, duplication reliability and sequence-number windows.

2 Prerequisite knowledge

You need the LTE radio protocol stack (Lecture 6), GTP-U and PFCP (Lecture 13), NFV data-plane techniques (Lecture 16), UPF topologies (Lecture 19) and the QoS model (Lecture 22). This lecture and the next one look at the 5G system "by plane": first the path of user data, then the signalling that controls it.

Key terms. The vocabulary of this lecture with its Persian equivalents:

Term	Persian	Meaning in this lecture
User plane	صفحهٔ کاربر	Functions and protocols that carry user data.
CU / DU / RU	واحد مرکزی / توزیع‌شده / رادیویی	Parts of a split gNB.
F1-U / N3 / N9	واسط‌های صفحهٔ کاربر	GTP-U interfaces CU–DU, RAN–UPF, UPF–UPF.
Functional split	تقسیم کارکردی	Point in the protocol stack where the RAN is divided.
Fronthaul	فرانت‌هال	Transport between DU and RU (or baseband and radio).
PDCP duplication	تکرار PDCP	Sending the same PDCP PDU over two legs for reliability.
SN window	پنجرهٔ شمارهٔ ترتیب	Range of sequence numbers in flight for reordering.
PDR / FAR / QER / URR	قواعد PFCP	Detection, forwarding, QoS and usage rules in the UPF.
DPDK / SmartNIC	شتاب صفحهٔ داده	Software and hardware acceleration of packet processing.
Local breakout	خروج محلی	Exiting to a data network close to the RAN.

3 Introduction and motivation

Everything else in a mobile network exists to serve the user plane: the path that carries the subscriber's packets. Its performance – throughput, latency, reliability, cost per bit – is what users experience and what operators pay for. In 5G the user plane is more flexible than ever: the gNB can be split into centralised and distributed units, the core user plane is a programmable UPF that can be placed anywhere from a central data centre to a factory, and several UPFs can be chained in one session.

This flexibility creates design questions that this lecture equips you to answer quantitatively: where to split the RAN given the available fronthaul, where to place UPFs given a latency budget, how to reach URLLC reliability with duplication, and how much the protocol headers cost [3]–[5].

4 Fundamental concepts

4.1 The end-to-end stack

User data pass (Figure 1): in the UE through SDAP, PDCP, RLC, MAC and PHY; over the radio to the gNB-DU (RLC, MAC, PHY); over F1-U in GTP-U to the gNB-CU-UP (PDCP, SDAP); over N3 in GTP-U to the UPF, possibly over N9 to further UPFs; and out of N6 to the data network. The radio layers have these roles [4]:

- **SDAP**: maps QoS flows to data radio bearers and marks QFIs (Lecture 22).
- **PDCP**: header compression (ROHC), ciphering and optional integrity protection, reordering and duplicate removal, PDCP duplication.
- **RLC**: segmentation and, in acknowledged mode, ARQ retransmission; NR's RLC no longer concatenates or reorders.
- **MAC**: multiplexing of logical channels, HARQ, scheduling.

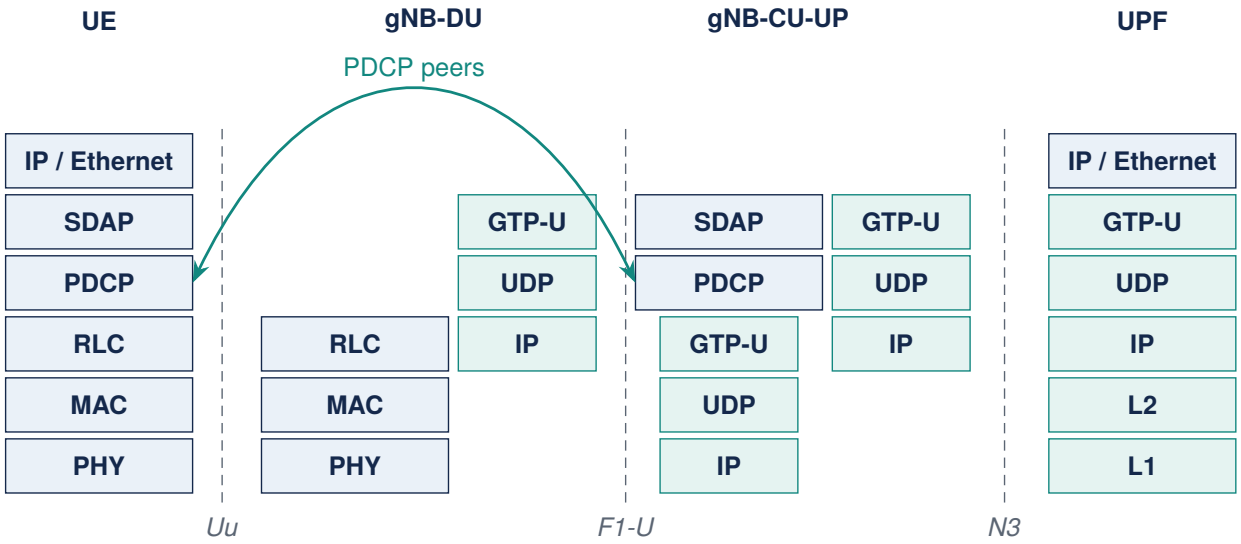


Figure 1. 5G user-plane protocol stack with a split gNB (CU-UP and DU).

4.2 Splitting the gNB

3GPP standardised a *higher-layer split* between a central unit (CU: RRC, SDAP, PDCP) and a distributed unit (DU: RLC, MAC, PHY), connected by F1 [3]; the CU is further split into control (CU-CP) and user plane (CU-UP), connected by E1. Lower-layer splits between the DU and a radio unit (RU) – in particular option 7.2, specified by the O-RAN Alliance – move part of the physical layer into the RU (Figure 2). The lower the split, the more functions can be centralised (simpler sites, coordination across cells), but the higher the fronthaul rate and the tighter the latency requirement (Section 6.2).

4.3 The UPF

The UPF processes every packet against the rules installed by the SMF (Figure 3) [2]: a *PDR* detects the packet (by TEID, UE address, SDF filter or QFI; the matching PDR with the highest precedence wins) and points to a *FAR* (forward with a new outer header, buffer, drop, duplicate), *QERs* (gating and rate limits) and *URRs* (counting and reporting volumes, time and events). Buffering for idle UEs follows a *BAR*. The UPF thus combines the S-GW and P-GW user-plane functions of

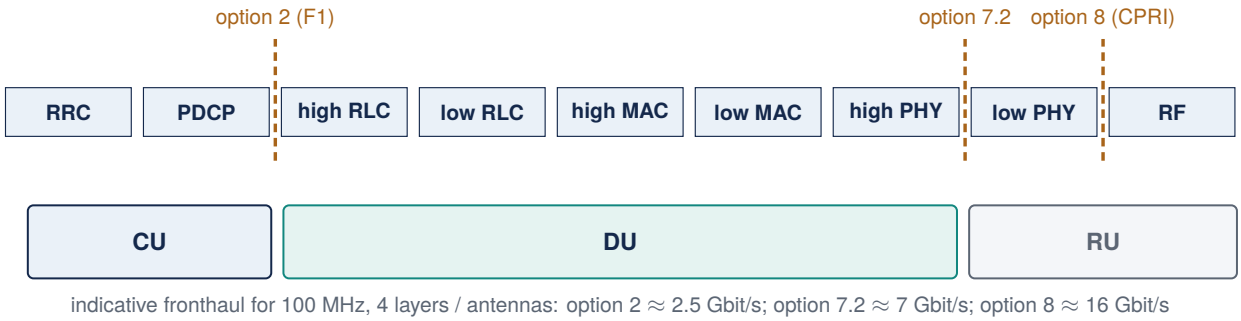


Figure 2. Selected RAN functional splits with indicative fronthaul rates (assumptions in Section 6.2).

the EPC (Lecture 13).

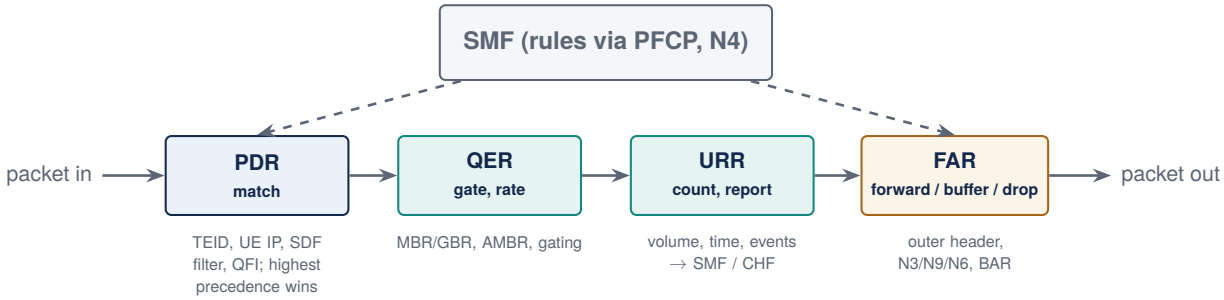


Figure 3. UPF packet processing with PFCP rules.

5 Detailed technical explanation

5.1 PDCP duplication and multi-connectivity

For URLLC, PDCP can duplicate packets and send the copies over two independent legs: two carriers of the same gNB (carrier aggregation) or two gNBs (dual connectivity). The receiver's PDCP discards the second copy. Duplication converts the independence of the legs into reliability (Section 6.4) and removes the interruption of a handover, at the cost of doubling the radio resources used for the duplicated flow. It can be activated dynamically by MAC control elements when the channel deteriorates.

5.2 Sequence numbers and reordering

PDCP numbers each SDU (12- or 18-bit SN); RLC in acknowledged mode numbers segments (12- or 18-bit SN), in unacknowledged mode 6- or 12-bit. The receiving PDCP delivers in order and waits up to a reordering timer for missing packets. The window of SNs that may be in flight is half the SN space; at high data rates a short SN wraps quickly, which is why NR introduced 18-bit SNs (Section 6.5).

5.3 F1-U and N3 transport

On F1-U the CU-UP and DU exchange PDCP PDUs in GTP-U with an NR user-plane protocol header that carries flow-control information (e.g. the DU's desired buffer size), so that the CU can pace the DU and avoid overflows when radio conditions change. On N3 and N9 the GTP-U header carries the QFI and the reflective QoS indicator in an extension header. Both run over IP transport networks that must preserve QoS by mapping QFIs/5QIs to DSCP values.

5.4 UPF implementation and placement

A UPF must classify, count and re-encapsulate every packet at line rate. Implementations range from software on servers with DPDK and SR-IOV (Lecture 16), through SmartNIC offload of encapsulation and classification, to programmable switches running P4 pipelines (Lecture 15) for the fast path while a CPU handles exceptions. Placement follows the latency budget (Section 6.3): central UPFs for Internet traffic, regional or edge UPFs for latency-sensitive and locally served traffic, on-premises UPFs for enterprise and industrial networks.

6 Mathematical models

6.1 Protocol overhead

Model. An IP packet of L bytes acquires SDAP, PDCP, RLC and MAC headers of $H = H_{\text{SDAP}} + H_{\text{PDCP}} + H_{\text{RLC}} + H_{\text{MAC}}$ bytes on the radio. The efficiency is

$$\eta = \frac{L}{L + H}. \tag{1}$$

Example. SDAP 1, PDCP 3 (18-bit SN), RLC AM 3, MAC subheader 3 bytes: $H = 10$ bytes. For $L = 1500$: $\eta = 99.3\,\%$; for a 60-byte voice packet (after ROHC compresses its 60-byte IP/UDP/RTP header to a few bytes, L is smaller still): $\eta = 85.7\,\%$. Integrity protection adds a 4-byte MAC-I to PDCP.

6.2 Fronthaul rate

Model. With split option 8 the fronthaul carries time-domain IQ samples: sampling rate f_s, b bits per I and per Q component, N_{ant} antennas and line coding c:

$$R_{\text{FH},8} = 2\, f_s\, b\, N_{\text{ant}}\, c. \tag{2}$$

With split 7.2 the fronthaul carries frequency-domain samples of the occupied resource elements per layer, approximately $R_{\text{FH},7.2} \approx 12\, N_{\text{PRB}}\, N_{\text{sym}}\, N_{\text{layers}}\, 2b'$ with N_{sym} symbols per second and compressed bit width b'.

Example. 100 MHz NR at 30 kHz: $f_s = 122.88$ Msample/s, $b = 16$, 4 antennas, $c = 66/64$: $R_{\text{FH},8} = 16.2$ Gbit/s; with 64 antennas 260 Gbit/s – impractical. Split 7.2 with 273 PRBs, 28 000 symbols/s, 4 layers, $b' = 9$: about 6.6 Gbit/s, scaling with layers rather than antennas. The higher-layer split (option 2) carries roughly the user data rate (about 2.5 Gbit/s for the carrier of Lecture 14).

Interpretation. Massive MIMO is only practical with splits that do beamforming in the RU; the choice of split is a trade-off between centralisation and transport cost.

6.3 Latency budget

Model. One-way user-plane latency between UE and an application server is the sum of its components:

$$T = T_{\text{radio}} + T_{\text{gNB}} + T_{\text{FH}} + T_{\text{BH}} + T_{\text{UPF}} + T_{\text{N6}}. \tag{3}$$

Example. With illustrative values (Figure 4): radio 2 ms, gNB processing 0.5 ms, fronthaul 0.1 ms, UPF 0.1 ms; an edge UPF with 0.2 ms transport and 0.1 ms to the server gives 3.0 ms, a central UPF 600 km away (3 ms transport) with 1 ms to the server gives 6.7 ms. Only the edge placement meets a 4 ms one-way target.

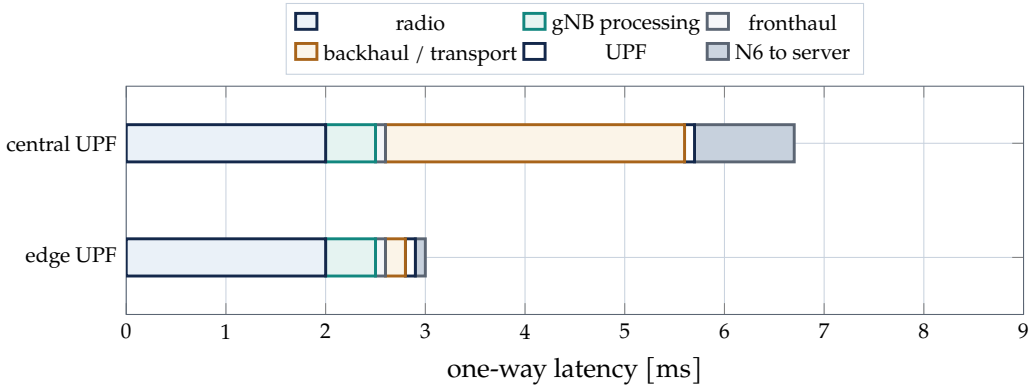


Figure 4. One-way user-plane latency budget with edge and central UPF (illustrative values, Eq. (3)).

6.4 Reliability with duplication

Model. If each of two independent legs loses a packet with probability p_1, p_2, duplication loses it only if both fail:

$$P_{\text{loss}} = p_1\, p_2. \tag{4}$$

Example. $p_1 = p_2 = 10^{-3}$ gives 10^{-6}, meeting a URLLC target that a single leg could reach only with a much more robust (and costly) MCS or more retransmissions – which might not fit the delay budget.

Limitation. Correlated losses (both carriers from the same site blocked by the same obstacle) break the independence assumption; dual connectivity with geographically separated nodes is more robust than carrier-level duplication.

6.5 Sequence-number window

Model. With an s-bit SN, at most 2^{s-1} PDUs may be in flight. At a packet rate P the window covers

$$T_{\text{win}} = \frac{2^{s-1}}{P}. \tag{5}$$

Example. 2 Gbit/s with 1500-byte packets: $P = 1.67 \times 10^5$ packets/s. A 12-bit SN covers 12.3 ms – shorter than HARQ and ARQ delays plus reordering; an 18-bit SN covers 0.79 s. High-throughput bearers therefore use 18-bit SNs.

7 Network architecture and diagrams

A typical 5G deployment combines RUs at the antenna sites, DUs at the cell sites or in nearby hubs (connected by 7.2 fronthaul), CU-UPs in regional data centres together with edge UPFs, and central UPFs for Internet traffic. Each element can be virtualised on common infrastructure (vRAN, Lecture 16), and the SMF chains UPFs per session (Figures 1 to 3).

8 Step-by-step procedures

Procedure 1 • Following a downlink packet

1. The packet arrives on N6 at the PSA UPF; a PDR matches it (UE address and SDF filter), the QER enforces rates, the URR counts it, and the FAR encapsulates it in GTP-U with the QFI towards the next hop (N9 or N3).
2. The CU-UP receives it on N3, SDAP maps the QFI to a DRB, PDCP compresses headers, ciphers and numbers it, and sends it in GTP-U over F1-U.
3. The DU's RLC segments it as scheduled by MAC; MAC multiplexes it into a transport block; PHY transmits it with HARQ.
4. The UE's layers reverse the process; PDCP reorders and delivers in sequence to SDAP and the IP layer.

Procedure 2 • Choosing a RAN split

1. Determine bandwidth, number of antennas and layers per cell and compute the fronthaul rate for candidate splits (Eq. (2)).
2. Compare with the available transport (fibre, microwave) and its latency; lower splits need sub-millisecond fronthaul latency.
3. Weigh the benefits of centralisation (coordination, pooling, simpler sites) against transport cost.
4. Choose the split per site type (e.g. 7.2 for urban massive MIMO, option 2 for remote sites with limited transport).

9 Worked examples

Worked example 1 • Overhead for small packets

Problem. IoT sensors send 20-byte payloads in IPv6/UDP without header compression. Compute the efficiency on the radio and with ROHC compressing IPv6/UDP to 4 bytes.
Solution. Without ROHC: $L = 20 + 48 = 68$, data efficiency $20/(68 + 10) = 25.6\,\%$. With ROHC: $L = 24$, $20/34 = 58.8\,\%$. Header compression matters most for small packets.

Worked example 2 • Fronthaul for massive MIMO

Problem. A 64-antenna RU with 100 MHz serves 16 layers. Compare option 8 and option 7.2 fronthaul (assumptions of Section 6.2).
Solution. Option 8: 260 Gbit/s. Option 7.2: $273 \times 12 \times 28\,000 \times 16 \times 18 = 26.4$ Gbit/s – ten times less, feasible over a 25 or 50 Gbit/s Ethernet link.

Worked example 3 • Duplication budget

Problem. Each leg has BLER 10^{-2} per transmission and can make 2 transmissions within the budget. What residual loss does a single leg reach, and what does duplication reach?
Solution. Single leg: $(10^{-2})^2 = 10^{-4}$. With duplication over two independent legs: $10^{-4} \times 10^{-4} = 10^{-8}$ (Eq. (4)) – beyond the 10^{-5} URLLC targets, leaving margin for correlation.

10 Technology comparison

Table 1. User plane in LTE/EPC and in NR/5GC.

	LTE / EPC	NR / 5GC
Radio layers	PDCP, RLC (with reordering), MAC	SDAP, PDCP, RLC (no reordering), MAC
RAN split	Monolithic eNB (CPRI to RRH)	CU/DU (F1), CU-CP/CU-UP (E1), lower-layer split 7.2
Core user plane	S-GW + P-GW (CUPS optional)	UPF chains (I-UPF, UL CL, PSA) controlled by PFCP
QoS on N3/S1-U	Tunnel per bearer	Tunnel per session, QFI per packet
Reliability	HARQ, ARQ	+ PDCP duplication, multi-connectivity
SN length	PDCP up to 18 bits (later)	PDCP 12/18, RLC AM 12/18 bits

Programmable all the way. From P4-programmed UPFs to virtualised CU/DU, the 5G user plane is software-defined end to end – which makes its performance a matter of engineering budgets (cycles per packet, microseconds per hop) rather than of fixed boxes.

11 Key takeaways

Key takeaways

- User data: UE (SDAP/PDCP/RLC/MAC/PHY) → DU (RLC/MAC/PHY) → F1-U → CU-UP (PDCP/SDAP) → N3 → UPF(s) → N6.
- Higher-layer split (option 2) needs about the user rate; option 8 scales with antennas ($2f_s b N_{\text{ant}} c$); 7.2 with layers.
- UPF processing: PDR match → QER → URR → FAR, programmed by the SMF over N4.
- Latency = $\sum$ components; edge UPFs remove milliseconds of transport.
- PDCP duplication gives $P_{\text{loss}} = p_1 p_2$; long SNs keep the window $2^{s-1}/P$ above reordering delays.

12 Exercises

Exercise 1 • Level 1 – conceptual • Which layer?

Which layer performs: (a) ciphering; (b) segmentation; (c) QoS flow to DRB mapping; (d) HARQ; (e) duplicate removal; (f) ARQ?

Exercise 2 • Level 1 – conceptual • Why split?

Give two advantages and two disadvantages of centralising the CU for many DUs.

Exercise 3 • Level 2 – numerical • Fronthaul

A 40 MHz carrier at 30 kHz is sampled at 61.44 Msample/s. Compute the option-8 fronthaul for 8 antennas with 16-bit samples and 66/64 coding. Can it run over a 25 Gbit/s link?

Exercise 4 • Level 2 – numerical • SN window

A bearer carries 5 Gbit/s of 1200-byte packets. Compute the window covered by 12-bit and 18-bit PDCP SNs. Which is needed if the reordering timer can reach 100 ms?

Exercise 5 • Level 3 – analytical • Placing UPFs

An operator has one central data centre (800 km from the farthest cells) and can build regional sites at most 150 km from any cell. A service needs 5 ms one-way latency with radio 2.5 ms, gNB 0.5 ms, UPF and server 0.4 ms. Which placement works, using 5 µs/km for transport? Discuss the cost.

Exercise 6 • Level 3 – analytical • Correlated duplication

Two legs each lose packets with probability 10^{-3}, but with probability q they are blocked together (common cause), in which case both lose. Model the loss as $P = q + (1-q)p^2$ with $p = 10^{-3}$ and find the largest q for which $P \le 10^{-5}$. What does this imply for deployment?

Solutions

Solution to Exercise 1

(a) PDCP; (b) RLC; (c) SDAP; (d) MAC; (e) PDCP; (f) RLC (acknowledged mode).

Solution to Exercise 2

Advantages: pooling of PDCP/RRC processing (statistical multiplexing), easier coordination and mobility between cells of the same CU, simpler and cheaper cell sites, software upgrades in one place. Disadvantages: transport between CU and DU needed (bandwidth and latency), larger failure domain (one CU outage affects many cells), and latency-sensitive functions cannot be centralised beyond a distance.

Solution to Exercise 3

$2 \times 61.44 \times 10^6 \times 16 \times 8 \times 66/64 = 16.2\,\text{Gbit/s}$: fits a 25 Gbit/s link for one carrier, but not two such carriers.

Solution to Exercise 4

$P = 5 \times 10^9/9600 = 5.2 \times 10^5$ packets/s. 12-bit: $2048/P = 3.9$ ms; 18-bit: $131\,072/P = 0.25$ s. With a 100 ms reordering timer, only the 18-bit SN is safe.

Solution to Exercise 5

Fixed part $2.5 + 0.5 + 0.4 = 3.4$ ms, leaving 1.6 ms of transport: $D \leq 1.6\ \text{ms}/5\ \mu\text{s/km} = 320$ km. The central site (800 km, 4 ms) fails; regional sites (150 km, 0.75 ms) work. Cost: more UPF sites (hardware, operations), regional peering or caching to serve content locally – to be justified by the revenue of latency-sensitive services.

Solution to Exercise 6

$q + (1-q)10^{-6} \leq 10^{-5} \Rightarrow q \leq 9 \times 10^{-6}$ (approximately). Common-cause blocking must be extremely rare: the legs should come from different sites and directions (dual connectivity), not just different carriers of the same antenna.

13 Further reading

The NR user-plane architecture is described in TS 38.300 [4] and the split RAN in TS 38.401 [3]; PFCP in TS 29.244 [2]; GTP-U in TS 29.281 [1]; UPF functions and topologies in TS 23.501 [5]. Dahlman et al. [7] explain the NR layer-2 protocols; Bosshart et al. [6] introduce the P4 language used in programmable UPFs.

References

[1] 3GPP, *General packet radio system (GPRS) tunnelling protocol user plane (GTPv1-U)*, TS 29.281, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/29281.htm.

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